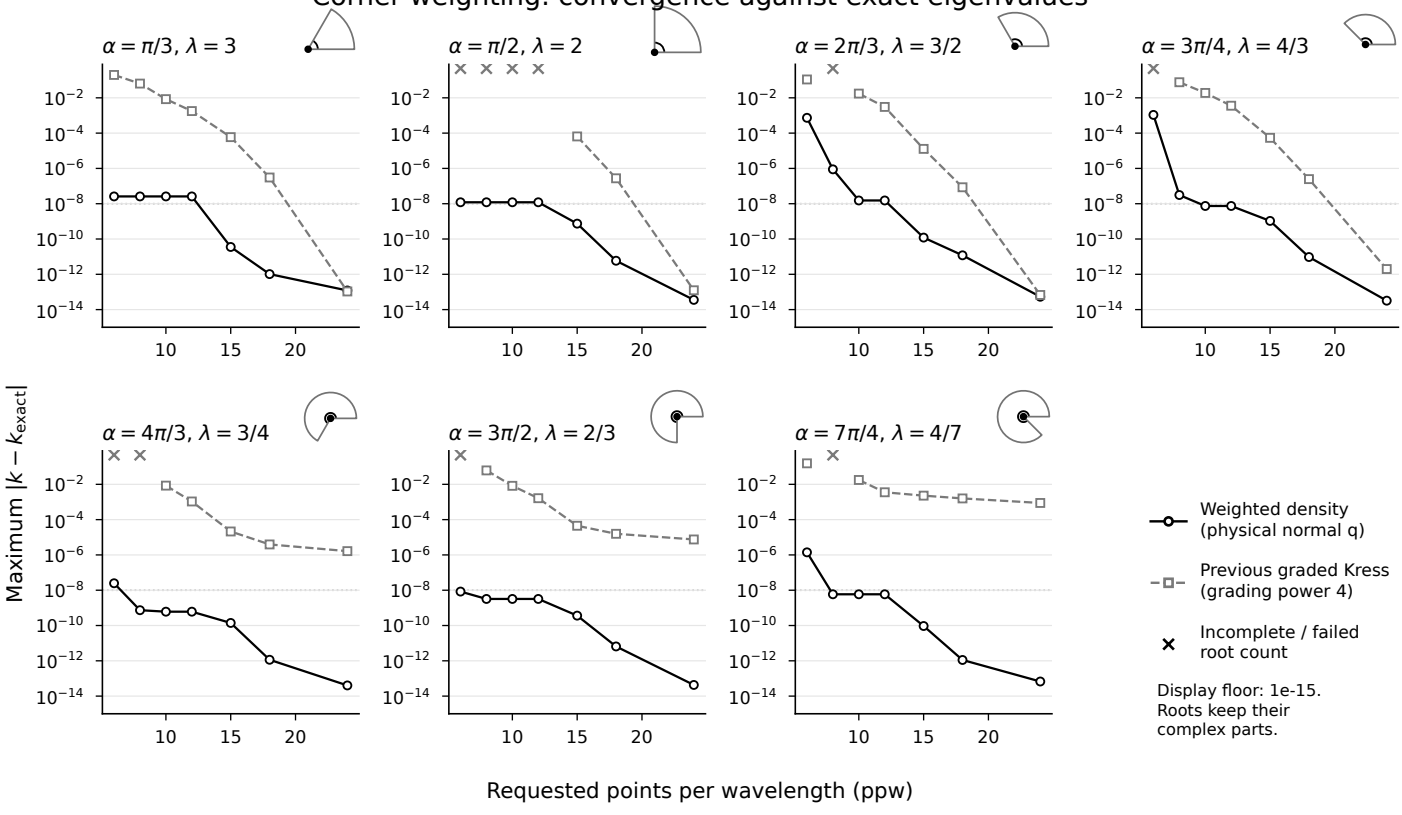

Error is max |k_computed - k_exact| over every root in the contour, retaining complex parts and multiplicities. Exact roots are zeros of J_(m*pi/alpha)(k), m >= 1. 10 solves have incomplete or failed root counts; crosses identify these and lines are broken there. At the 1e-8 target, the measured weighted-node saving spans 30% to 52% among 4 angles where both methods reach the target. Radius-1 Dirichlet sectors have two additional pi/2 joins. Insets mark the varied corner. Weighting factors out the leading q(r) ~ r^(lambda-1) power, lambda = pi/alpha; higher powers and possible logarithms remain, so exponential convergence is not assumed.